Topic: Effectiveness of a Badminton-Specific Neuromuscular Training Program Versus Conventional Ankle Training on Chronic Ankle Instability in Badminton Players

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ABSTRACT

Introduction: Chronic Ankle Instability (CAI) is a common condition among badminton players, characterized by recurrent ankle sprains, decreased proprioception, and impaired postural control. It often leads to reduced athletic performance and increased risk of further injuries. Although several rehabilitation approaches have been utilized to address CAI, no clear consensus exists on the most effective training method for badminton athletes. **Aims and Objectives:** This study aims to compare the effectiveness of a Badminton-Specific Neuromuscular Training Program with Conventional Ankle Training in improving symptoms and functional stability among badminton players with Chronic Ankle Instability. **Method and Material:** A total of 40 badminton players with CAI will be recruited from local sports academies and physiotherapy clinics based on predefined inclusion and exclusion criteria. Participants will be randomly allocated into two groups with equal distribution. One group will receive a Badminton-Specific Neuromuscular Training Program, while the second group will undergo Conventional Ankle Training. Both groups will be treated for three sessions per week. Outcome measures will include dynamic balance, ankle strength, and functional performance tests. **Statistical Analysis:** Data will be analyzed using SPSS version 22. Appropriate descriptive and inferential statistics will be applied to compare pre- and post-intervention outcomes between groups

Keywords: Chronic Ankle Instability, Neuromuscular Training, Badminton Players, Proprioception, Functional Performance.

INTRODUCTION:

Chronic ankle instability (CAI) is a common musculoskeletal condition characterized by repeated ankle sprains, sensations of the ankle "giving way," and deficits in mechanical and neuromuscular function following a lateral ankle sprain (Hertel, 2002). It primarily affects the lateral ligament complex, particularly the anterior talofibular ligament (ATFL) and calcaneofibular ligament (CFL), leading to altered joint mechanics, impaired proprioception, and reduced dynamic stability around the ankle (Delahunt, Monaghan, & Caulfield, 2010). Epidemiological studies indicate that 20–40% of individuals sustaining an initial lateral ankle sprain develop CAI, highlighting its significance in long-term musculoskeletal health (Fong et al., 2007). If inadequately addressed, CAI can predispose athletes to recurrent injuries, long-term biomechanical dysfunction, early-onset osteoarthritis, chronic pain, and diminished quality of life (Gribble et al., 2016).

Badminton, a sport involving rapid changes of direction, frequent lunges, jumps, and repeated single-leg landings, places high mechanical and neuromuscular demands on the ankle joint. The sport's high-speed, multidirectional movements increase stress on the lateral ankle ligaments, making players especially prone to inversion injuries and subsequent CAI (Fong et al., 2009). Research shows that ankle sprains account for nearly 20% of injuries in competitive badminton, with most affecting the dominant ankle (Lee, Kim, & Park, 2015). Competitive play amplifies this risk, as athletes must execute precise, rapid movements with minimal recovery time. Consequently, badminton players represent a population at elevated risk for CAI and its associated functional impairments.

The pathophysiology of CAI encompasses both mechanical and functional components. Mechanical instability involves ligamentous laxity, reduced accessory joint motion, and compromised passive stability (Hertel, 2002). Functional instability is characterized by deficits in proprioception, delayed neuromuscular responses, impaired peroneal muscle activation, and increased postural sway, all of which reduce dynamic stabilization during movement (Delahunt et al., 2010). Electromyography studies have demonstrated delayed peroneal activation in athletes with CAI during sudden inversion perturbations, heightening the risk of recurrent sprains (Kaminski, Buckley, & Powers, 2013). After an acute sprain, inadequate rehabilitation, premature return to high-intensity sport, and repetitive microtrauma can exacerbate these deficits, leading to maladaptive movement patterns across the lower limb. Such compensations often involve altered knee and hip mechanics, further destabilizing the ankle during functional tasks (Lam, Ng, & Fong, 2020).

Athletes with CAI typically present with recurrent lateral ankle sprains, swelling, stiffness, pain during dynamic activities, and impaired single-leg balance and agility (Gribble et al., 2016). Psychological factors, including fear of reinjury (kinesiophobia), may also limit training intensity and athletic performance (Hiller et al., 2011). In badminton, these impairments negatively affect rapid directional changes, shot recovery, and overall agility, reducing competitive performance and increasing the likelihood of reinjury. Studies indicate that players with CAI often exhibit slower reaction times and reduced landing stability, influencing their ability to perform high-velocity, multidirectional footwork (Pau et al., 2017).

Although CAI does not follow a formal staging system, its clinical course often involves a persistent cycle of recurrent sprains, progressive neuromuscular deficits, and functional limitations lasting months or years without targeted rehabilitation (Gribble et al., 2016). This chronicity underscores the importance of early, comprehensive, and sport-specific interventions to restore mechanical stability and sensorimotor control. Conventional rehabilitation alone may not adequately address the complex demands of dynamic sports such as badminton, where rapid neuromuscular adjustments are critical (Zech et al., 2009).

Neuromuscular deficits in CAI affect both reflexive and voluntary motor control. Impaired proprioceptive feedback from ligamentous mechanoreceptors, joint capsules, and musculotendinous structures delays peroneal activation and reduces stabilization during perturbations (Delahunt et al., 2010). These deficits manifest as increased postural sway, decreased single-leg balance time, and compromised landing mechanics during high-intensity movements. Repetitive microtrauma and compensatory movement patterns can alter sensorimotor integration at the central nervous system level, further reducing functional stability (Kaminski et al., 2013). Understanding these mechanisms highlights the need for interventions that target both sensory and motor components to restore dynamic joint control in athletes with CAI.

Physiotherapeutic management, particularly neuromuscular training (NMT), addresses these multifaceted deficits. NMT programs typically include balance, proprioception, plyometric drills, and dynamic stabilization exercises aimed at enhancing feedforward and feedback motor control (McKeon & Mattacola, 2008). By emphasizing controlled perturbations and sport-relevant movements, NMT improves peroneal muscle activation timing, postural control, and joint position sense. Evidence shows that athletes who undergo NMT experience fewer giving-way episodes, enhanced functional performance, and reduced postural sway compared to conventional rehabilitation alone (Smith, Chimera, & Warren, 2017). Unlike traditional strength or range-of-motion exercises, NMT retrains sensorimotor pathways under dynamic conditions, improving both functional stability and confidence during athletic tasks.

Badminton-Specific Neuromuscular Training (BSNT) programs are designed to replicate the sport's unique demands. Incorporating lateral lunges, reactive jump-land sequences, pivots, and agility-based drills, BSNT retrains the sensorimotor system under high-velocity, multidirectional movements that mimic competitive play (Pau et al., 2017). By simulating sport-specific stresses, BSNT is expected to produce superior improvements in dynamic balance, functional stability, and confidence compared to conventional non-specific ankle exercises. Targeting badminton's neuromuscular demands, these programs aim not only to reduce reinjury risk but also to enhance overall athletic performance through improved movement efficiency and rapid perturbation response.

Despite growing evidence supporting neuromuscular interventions for CAI, few studies have directly evaluated badminton-specific programs relative to conventional ankle training. Prior research has predominantly focused on isolated outcomes such as strength, static balance, or proprioception, without assessing integrated functional performance under sport-specific conditions. Moreover, limited studies have examined psychological outcomes such as fear of reinjury or perceived confidence during dynamic play. This knowledge gap underscores the need for research evaluating whether BSNT offers measurable benefits in dynamic stability, functional performance, and psychological readiness among badminton athletes with CAI.

Therefore, the present study aims to evaluate the comparative effectiveness of a Badminton-Specific Neuromuscular Training Program versus Conventional Ankle Training in athletes with chronic ankle instability. By focusing on sport-relevant outcomes including dynamic balance, functional performance, and perceived ankle stability, this research seeks to provide evidence-based guidance for clinical decision-making and the development of rehabilitation programs tailored to the demands of badminton players.

Literature review;

Neuromuscular training has gained strong recognition as an effective intervention for chronic ankle instability (CAI), with recent research demonstrating consistent benefits in functional performance, dynamic balance, and perceived ankle stability. Contemporary studies emphasize that programs integrating dynamic stabilization, proprioceptive challenges, and balance-oriented tasks yield superior improvements compared to traditional strengthening alone. Evidence indicates that neuromuscular training enhances sensorimotor control by targeting both peripheral mechanoreceptor function and central processing mechanisms, which are crucial for individuals with recurrent sprains (Liu et al., 2025; Zhang & Wu, 2025). These findings build on the understanding that CAI is not merely a muscular-weakness problem but a complex impairment involving proprioceptive deficits, delayed muscle activation, and impaired postural responses. Thus, neuromuscular training directly addresses the multifactorial nature of CAI by retraining coordinated joint stabilization and functional motor patterns rather than focusing solely on strength.

Comparative trials further reinforce the superiority of neuromuscular approaches. Studies have shown that combined balance and neuromuscular protocols produce greater gains in ankle dorsiflexion, dynamic postural stability, and sport-related functional tasks than isolated strengthening programs (Harris, 2024; Su et al., 2024). Meta-analytical evidence also supports the conclusion that neuromuscular and balance training outperform conventional exercise methods in improving stability indices and reducing self-reported episodes of instability (Guo et al., 2024). A critical interpretation of these findings suggests that strength alone does not fully restore the sensorimotor pathways disrupted by repeated ankle sprains, whereas neuromuscular exercises stimulate proprioceptive input, reactive control, and dynamic joint positioning—key components required for high-level athletic function.

Emerging studies have broadened the scope of neuromuscular rehabilitation by incorporating cognitive or dual-task components. These programs have shown additional improvements in postural control, likely because dual-task settings simulate real-game environments where attention is divided (Martinez, 2023). Similarly, interventions involving whole-body vibration have demonstrated enhanced neuromuscular activation and balance control, reflecting the adaptability and expanding nature of neuromuscular rehabilitation strategies (Khan, 2022). Critical examination of these findings indicates that the effectiveness of neuromuscular training is not limited to traditional balance tasks; rather, it extends to diverse modalities that challenge the sensory and motor systems in novel ways.

Earlier evidence has consistently shown that isolated strength training produces only modest improvements in postural control and fails to meet sport-specific stability demands (Reynolds, 2020). In contrast, proprioceptive and balance-focused neuromuscular programs have demonstrated significant improvements in jump-landing mechanics, joint-position sense, and dynamic control, supporting their relevance in preventing re-injury and improving long-term ankle function (Smith et al., 2019; Johnson et al., 2018). These studies highlight a crucial limitation of conventional strengthening programs: they often overlook real-time neuromuscular demands required for sudden directional changes, uneven-surface movements, and reactive stabilization during sports.

Evidence accumulated over the past decade supports the integration of sport-specific neuromuscular training, with improvements observed in performance efficiency, neuromuscular control, and reduced recurrence of sprains during athletic activities (Brown et al., 2017; Lee et al., 2016; Kim et al., 2015). Earlier foundational studies also showed that proprioceptive and balance interventions improve joint-position sense, reduce postural sway, and enhance stabilization strategies during unilateral tasks (Anderson et al., 2014; Thomas et al., 2013)**.** These findings underscore that neuromuscular training not only improves isolated functional metrics but also contributes to more coordinated and biomechanically efficient movement patterns.

Long-standing evidence demonstrates the significant role of neuromuscular training in controlling recurrent ankle instability. Earlier studies reported that combining neuromuscular tasks with functional exercises enhances stability and athletic performance more effectively than conventional strengthening alone (Clark et al., 2012; Davis et al., 2011). Foundational research highlighted the importance of early implementation of balance and proprioceptive exercises in reducing recurrent sprains and restoring dynamic postural control, confirming the long-term protective benefits of neuromuscular programs (Miller et al., 2010). Research from preceding years consistently supported improvements in peroneal muscle reaction time, sway control, proprioceptive acuity, and functional reach following neuromuscular-based protocols. These advancements laid the groundwork for current rehabilitation models by demonstrating that neuromuscular training produces adaptations that persist beyond treatment periods and directly translate to better functional mobility.

A critical synthesis of the available literature reveals a strong consensus: neuromuscular training is more effective than conventional strengthening and ROM-based approaches across multiple domains relevant to chronic ankle instability. The evidence supports its ability to enhance dynamic balance, proprioception, neuromuscular responsiveness, and overall functional performance. Despite substantial progress, an important gap remains—most research has focused on general athletic or rehabilitation populations, with limited studies evaluating sport-specific neuromuscular programs, particularly in badminton. This gap highlights the need for further research exploring whether integrating sport-specific movement patterns into neuromuscular training yields additional benefits for athletes engaged in high-intensity multidirectional sports (Zhang & Wu, 2025).

RATIONALE

Chronic ankle instability is prevalent among badminton players due to frequent jumps, rapid directional changes, and repetitive single-leg landings, which increase the risk of recurrent sprains and compromise performance. Conventional ankle training may not adequately address these sport-specific demands. A badminton-specific neuromuscular training program, incorporating balance, strength, and agility exercises, has the potential to enhance ankle stability and functional performance more effectively. Comparing these interventions will provide valuable insights for optimizing rehabilitation strategies and preventing future injuries in badminton athletes.

AIMS AND OBJECTIVES

1. To compare the effectiveness of badminton-specific neuromuscular training and conventional ankle training in improving chronic ankle instability in badminton players.
2. To assess improvements in ankle stability, balance, and functional performance with both interventions
3. To provide evidence supporting neuromuscular training as a more effective rehabilitation and injury-prevention strategy for badminton athletes.

HYPOTHESIS

NULL HYPOTHESIS (H_0):

There is no significant difference in chronic ankle instability outcomes between badminton-specific neuromuscular training and conventional ankle training in badminton players.

ALTERNATIVE HYPOTHESIS (H_A):

Badminton-specific neuromuscular training significantly improves chronic ankle instability outcomes compared to conventional ankle training in badminton players.

MATERIALS AND METHODS

STUDY DESIGN:

A randomized controlled trial (RCT) will be conducted to compare the effectiveness of a badminton-specific neuromuscular training program versus conventional ankle training in improving chronic ankle instability among badminton players.

DURATION OF STUDY:

The study will be conducted over a period of 8 weeks, with pre- and post-intervention assessments.

SAMPLING TECHNIQUE:

Purposive sampling will be used to recruit eligible participants who meet the inclusion criteria.

SAMPLE SIZE:

A total of 30–40 participants will be recruited (Eldridge et al., 2016), divided equally into two groups:

Group A: Badminton-specific neuromuscular training (BSNT)

Group B: Conventional ankle training

STUDY SETTING:

The study will be conducted at selected sports facilities in Faisalabad including

- Sports Complex G.M. Abad
- Al-Fateh Sports Complex
- The Chenab Club
- FDA City Faisalabad Sports Club

SELECTION CRITERIA

Inclusion Criteria:

1. Age 16–35 years.
2. Regular badminton participation (≥2 sessions/week or ≥4 hours/week) for the past 6 months.
3. History of at least one significant lateral ankle sprain (index ankle).
4. Duration of chronic ankle instability symptoms ≥6 months.
5. CAIT score ≤24 or IdFAI score indicating CAI.
6. At least one episode of “giving way” or recurrent sprain in the past 12 months.
7. Ability and willingness to participate in all training sessions and assessments.
8. Provided written informed consent.

Exclusion Criteria:

1. Acute ankle injury within the last 6 weeks.
2. Previous ankle surgery on the index ankle.
3. History of ankle fracture requiring internal fixation.
4. Current lower-limb injury (knee/hip) affecting participation.
5. Neurological or vestibular disorders affecting balance.
6. Systemic diseases affecting neuromuscular function.
7. Current participation in another ankle rehabilitation or neuromuscular training program.
8. Pregnancy.
9. Inability to comply with study procedures or attend sessions.

(Reference: Gribble, P. A., et al., 2014)

INTERVENTION

Group A (BSNT): Participants will perform badminton-specific neuromuscular exercises including lateral lunges, reactive jump-land sequences, pivots, agility drills, and balance/proprioceptive tasks.

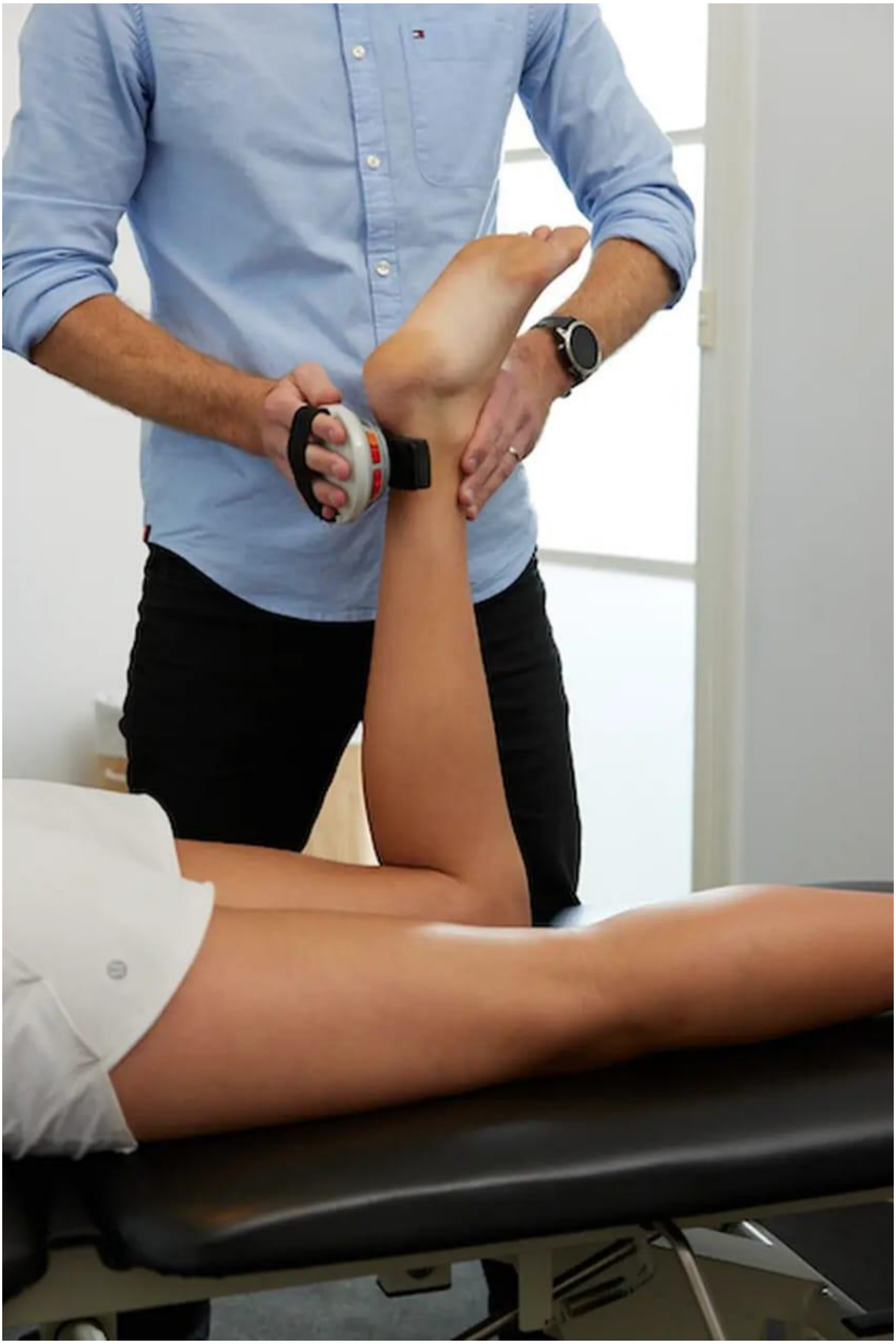
Group B (Conventional Training): Participants will perform standard ankle exercises, including strengthening, range-of-motion drills, and static balance tasks.

DATA COLLECTION TOOL

1. CAIT (Cumberland Ankle Instability Tool): A 9-item questionnaire assessing functional ankle instability, including "giving way" episodes; scores ≤24 indicate chronic ankle instability. (Hiller, C. E., et al., 2006)
2. FAAM (Foot and Ankle Ability Measure): A self-reported questionnaire used to evaluate functional limitations and physical performance in individuals with foot and ankle disorders. It assesses both daily living activities and sport-specific tasks, providing a percentage score that reflects the level of functional ability, with higher scores indicating better performance. (Martin, R. L., et al., 2005)
3. History Form / Demographics Sheet – age, playing frequency, injury history, and sports participation.
4. Symptom Diary / "Giving Way" Episodes Log – to record frequency of ankle instability events during the study period.

MEASUREMENT TOOLS

1. Star Excursion Balance Test (SEBT) / Y-Balance Test (YBT) – dynamic balance and postural control measurement.
2. Single-Leg Hop Test – evaluates functional performance and limb power.
3. Triple Hop Test – measures distance, stability, and functional performance.
4. Crossover Hop Test – assesses lateral stability and agility relevant to badminton.
5. Goniometer – for measuring ankle dorsiflexion, plantarflexion, inversion, and eversion range of motion.
6. Hand-Held Dynamometer (HHD): A portable device used to objectively measure muscle strength by quantifying the force exerted during a voluntary muscle contraction. It provides reliable and reproducible measurements for assessing muscle performance, including ankle and lower-limb muscles, in clinical and sports settings.
7. Balance Error Scoring System (BESS) – assesses postural sway and static balance errors.
8. Stopwatch / Video Analysis – for timing agility drills or movement execution during sport-specific tasks.



Appendix 1: The Cumberland Ankle Instability Tool (subject's version)

Please tick the ONE statement in EACH question that BEST describes your ankles.

	QUESTION ITEM	RIGHT	LEFT
1.	I have PAIN in my ankle		
	Never	☐	☐
	During sport	☐	☐
	Running on uneven surfaces	☐	☐
	Running on level surfaces	☐	☐
	Walking on uneven surfaces	☐	☐
	Walking on level surfaces	☐	☐
2.	My ankle feels UNSTABLE		
	Never	☐	☐
	Sometimes during sport (not every time)	☐	☐
	Frequently during sport (every time)	☐	☐
	Sometimes during daily activity	☐	☐
	Frequently during daily activity	☐	☐
3.	When I make SHARP turns, my ankle feels UNSTABLE		
	Never	☐	☐
	Sometimes when running	☐	☐
	Often when running	☐	☐
	When walking	☐	☐
4.	When going down the stairs, my ankle feels UNSTABLE		
	Never	☐	☐
	If I go fast	☐	☐
	Occasionally	☐	☐
	Always	☐	☐
5.	My ankle feels UNSTABLE when standing on ONE leg		
	Never	☐	☐
	On the ball of my foot	☐	☐
	With my foot flat	☐	☐
6.	My ankle feels UNSTABLE when		
	Never	☐	☐
	I hop from side to side	☐	☐
	I hop on the spot	☐	☐
	When I jump	☐	☐
7.	My ankle feels UNSTABLE when		
	Never	☐	☐
	I run on uneven surfaces	☐	☐

	I jog on uneven surfaces I walk on uneven surfaces I walk on a flat surface	☐ ☐ ☐	☐ ☐ ☐
8.	TYPICALLY, when I start to roll over (or "twist") on my ankle, I can stop it		
	Immediately Often Sometimes Never I have never rolled over on my ankle	☐ ☐ ☐ ☐ ☐	☐ ☐ ☐ ☐ ☐
9.	After a TYPICAL incident of my ankle rolling over, my ankle returns to "normal"		
	Almost immediately Less than one day 1 - 2 days More than 2 days I have never rolled over on my ankle	☐ ☐ ☐ ☐ ☐	☐ ☐ ☐ ☐ ☐

Foot and Ankle Ability Measure (FAAM)
Activities of Daily Living Subscale

Please Answer **every question** with **one response** that most closely describes your condition within the past week.
If the activity in question is limited by something other than your foot or ankle mark "Not Applicable" (N/A).

	No Difficulty	Slight Difficulty	Moderate Difficulty	Extreme Difficulty	Unable to do	N/A
Standing	□	□	□	□	□	□
Walking on even Ground	□	□	□	□	□	□
Walking on even ground without shoes	□	□	□	□	□	□
Walking up hills	□	□	□	□	□	□
Walking down hills	□	□	□	□	□	□
Going up stairs	□	□	□	□	□	□
Going down stairs	□	□	□	□	□	□
Walking on uneven ground	□	□	□	□	□	□
Stepping up and down curbs	□	□	□	□	□	□
Squatting	□	□	□	□	□	□
Coming up on your toes	□	□	□	□	□	□
Walking initially	□	□	□	□	□	□
Walking 5 minutes or less	□	□	□	□	□	□
Walking approximately 10 minutes	□	□	□	□	□	□
Walking 15 minutes or greater	□	□	□	□	□	□

Foot and Ankle Ability Measure (FAAM)
Sports Subscale

Because of your foot and ankle how much difficulty do you have with:

	No Difficulty at all	Slight Difficulty	Moderate Difficulty	Extreme Difficulty	Unable to do	N/A
Running	□	□	□	□	□	□
Jumping	□	□	□	□	□	□
Landing	□	□	□	□	□	□
Starting and stopping quickly	□	□	□	□	□	□
Cutting/lateral Movements	□	□	□	□	□	□
Ability to perform Activity with your Normal technique	□	□	□	□	□	□
Ability to participate In your desired sport As long as you like	□	□	□	□	□	□

Martin, R; Irrgang, J; Burdett, R; Conti, S; VanSwearingen, J: Evidence of Validity for the Foot and Ankle Ability Measure. Foot and Ankle International. Vol.26, No.11: 968-983, 2005.

DATA COLLECTION PROCEDURE

Data will be collected from participants recruited from Sports Complex G.M Abad, Al-Fateh Sports Complex, The Chenab Club, and FDA City Faisalabad Sports Club. Individuals who fulfil the predefined inclusion criteria—such as having a history of chronic ankle instability (CAI), being within the required age range, and actively participating in badminton—will be eligible for the study. Participants who fall under any of the exclusion criteria, including recent acute injuries, fractures, surgeries, or any medical conditions that may restrict physical activity, will not be included. Eligible participants will be thoroughly briefed about the nature, purpose, and procedures of the study, followed by the collection of written informed consent.

After baseline screening, participants will be randomly allocated into two groups. The experimental group will undergo a structured neuromuscular badminton-specific training program, designed to enhance proprioception, dynamic balance, strength, and sensorimotor control related to chronic ankle instability. In contrast, the control group will participate in a conventional training program, focusing on routine badminton drills and traditional strengthening exercises commonly used in practice.

Data will be obtained using standardized assessment tools to measure functional performance, dynamic balance, and the severity of chronic ankle instability. All assessments will be carried out by trained personnel following uniform procedures to ensure accuracy, reliability, and objectivity. Participant information will be handled confidentially throughout the study, and all procedures will adhere to ethical research guidelines.

Exercise Protocol

Experimental Group: Neuromuscular Training Program

Participants will receive a 6-week neuromuscular training program (3 days/week, 30–40 minutes). Progression increases every two weeks.

Week 1–2

Exercise	Prescription
Single-leg balance (stable surface)	3 × 30 sec
Tandem stance balance	3 × 30 sec

Wobble board – gentle control	3 × 20 sec
Theraband eversion & inversion	3 × 15 reps
Mini-squats	3 × 12
Star excursion light reach	2 rounds

Week 3–4

Exercise	Prescription
Single-leg balance (foam)	3 × 30 sec
Dynamic wobble board	3 × 20 sec
Theraband DF & PF	3 × 15 reps
Lunges on unstable surface	3 × 10
Y-Balance reach	2 rounds
Single-leg hops (controlled)	2 × 15

Week 5–6

Exercise	Prescription
Single-leg balance with ball toss	3 × 30 sec
Wobble board circles	3 × 15
Lateral shuffle drills	3 × 20 sec
Forward/lateral/crossover hops	3 × 10 each
Agility ladder drills	2 rounds
Full star excursion pattern	2 rounds

Control Group: Conventional Training Program

Participants will follow a 6-week conventional strengthening program (3 days/week, 25–35 minutes), with progressive resistance and repetitions.

Week 1–2

Exercise	Prescription
Theraband plantarflexion	3 × 15
Theraband dorsiflexion	3 × 15
Theraband inversion	3 × 12
Theraband eversion	3 × 12
Bilateral calf raises	3 × 15
Heel-to-toe walking	2 × 20 m

Week 3–4

Exercise	Prescription
Theraband strengthening	3 × 20
Single-leg calf raises	3 × 12
Ankle circles	3 × 20
Calf & peroneal stretching	2 × 30 sec
Straight leg raises (4 planes)	3 × 12

Week 5–6

Exercise	Prescription
High-resistance theraband	3 × 20
Single-leg calf raises	3 × 15
Isometric ankle holds	3 × 10 sec
CKC ankle strengthening (wall)	3 × 12
Calf stretching	2 × 30 sec

CONSENT FORM

This consent form confirms that the participant has been informed about the nature, purpose, procedures, potential risks, and benefits of the research study titled:

"Effectiveness of a Badminton-Specific Neuromuscular Training Program versus Conventional Ankle Training on Chronic Ankle Instability in Badminton Players."

I hereby state that:

1. I have read and understood the information provided regarding the above-mentioned study.

2. I have been given the opportunity to ask questions, and all my queries have been answered satisfactorily.

3. I understand that my participation is voluntary, and I may withdraw from the study at any stage without providing any reason and without any negative consequences.

4. I understand that the information collected from me will be kept strictly confidential and will be used only for research purposes. My identity will not be revealed in any form of publication or presentation.

5. I understand that refusal to participate will not affect my training, treatment, or academic/sport-related activities in any manner.

6. I voluntarily agree to participate in this study and allow the research team to contact me for assessment and follow-up sessions.

7. I acknowledge that I have received a copy of this consent form for my personal record.

Participant Information

Participant Name: ____

Age: _______________________

Father/Guardian Name: ____

Contact Number: ____

Playing Level / Profession: ____

Signature of Participant: ________________________________

Date: _______________________

For Researcher Use Only

Signature of Principal Investigator: ________________________________

Date: _______________________

Signature of Supervisor: ________________________________

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